MINI PROJECT

ON

ONE-WAY ANOVA USING SPSS SOFTWARE

(DATA ANALYSIS FOR BUSINESS DECISIONS-24BAP-733)

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE IN

MASTERS OF BUSINESS ADMINISTRATION (MBA)

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One-way ANOVA in SPSS Statistics

Introduction

The One-way analysis of variance (ANOVA) is used to determine whether there are any statistically significant differences between the means of two or more independent (unrelated) groups. For example, you could use a One-way ANOVA to understand whether exam performance differed based on test anxiety levels amongst students, dividing students into three independent groups (e.g., low, medium and high-stressed students). Also, it is important to realize that the one-way ANOVA is an **omnibus** test statistic and cannot tell you which specific groups were statistically significantly different from each other; it only tells you that at least two groups were different. Since you may have three, four, five or more groups in your study design, determining which of these groups differ from each other is important.

Importance of Using SPSS for One-Way ANOVA

1. **Automates Calculations**
   - One-Way ANOVA involves calculating means, variances, sums of squares, F-values, and p-values.
   - Doing this manually is time-consuming and prone to errors.
   - SPSS calculates all these accurately in seconds.
2. **Handles Large Data Easily**

   For small datasets, manual calculation is possible, but with large datasets, SPSS efficiently handles hundreds or thousands of entries without mistakes.

3. **Provides Descriptive Statistics**

   SPSS gives means, standard deviations, and counts for each group, which helps in understanding the data distribution before testing significance.

4. **Generates ANOVA Table Automatically**
    - Produces a clear ANOVA table including Between-Groups, Within-Groups, F-value, and p-value.
    - Makes interpretation straightforward without manual formula work.

5. **Reduces Human Error**

    Manual calculations of sum of squares, degrees of freedom, and mean squares are error-prone. SPSS ensures accuracy and reliability.

6. **Saves Time**

    What may take hours manually can be done in minutes with SPSS, letting you focus on interpretation rather than calculation.

Assumptions

To make sure the results of a one-way ANOVA are valid, certain assumptions must be satisfied:

1. **Dependent variable is continuous**
    - The variable being measured (e.g., completion time, marks, weight) should be on an interval or ratio scale.
2. **Independent variable is categorical**
    - The factor (e.g., type of training, group, treatment) should consist of two or more independent groups.
3. **Independence of observations**
    - Each participant belongs to only one group.
    - Observations are independent of each other (no repeated measures, no overlap).
4. **No significant outliers**
    - The dependent variable for each group should not contain extreme outlier values, as these can distort the ANOVA results.

5. **Normality**
    - The dependent variable should be approximately **normally distributed** within each group.
6. **Homogeneity of variances**
    - The variance of the dependent variable should be roughly **equal across groups**.

Example

A manager wants to raise the productivity at his company by increasing the speed at which his employees can use a particular spreadsheet program. As he does not have the skills in-house, he employs an external agency which provides training in this spreadsheet program. They offer 3 courses: a beginner, intermediate and advanced course. He is unsure which course is needed for the type of work they do at his company, so he sends 10 employees on the beginner course, 10 on the intermediate and 10 on the advanced course. When they all return from the training, he gives them a problem to solve using the spreadsheet program, and times how long it takes them to complete the problem. He then compares the three courses (beginner, intermediate, advanced) to see if there are any differences in the average time it took to complete the problem.

Employee	Beginner	Intermediate	Advanced
1	22	18	15
2	25	21	17
3	24	20	14
4	28	26	20
5	27	24	18
6	28	24	16
7	27	23	19
8	29	19	17
9	29	22	21
10	33	20	15

Hypothesis

H_0: There is **no significant difference in the average time taken** to solve the spreadsheet problem among employees attending the Beginner, Intermediate, and Advanced courses.

H_1: There is **significant difference in the average time taken** to solve the spreadsheet problem among employees attending the Beginner, Intermediate, and Advanced courses.

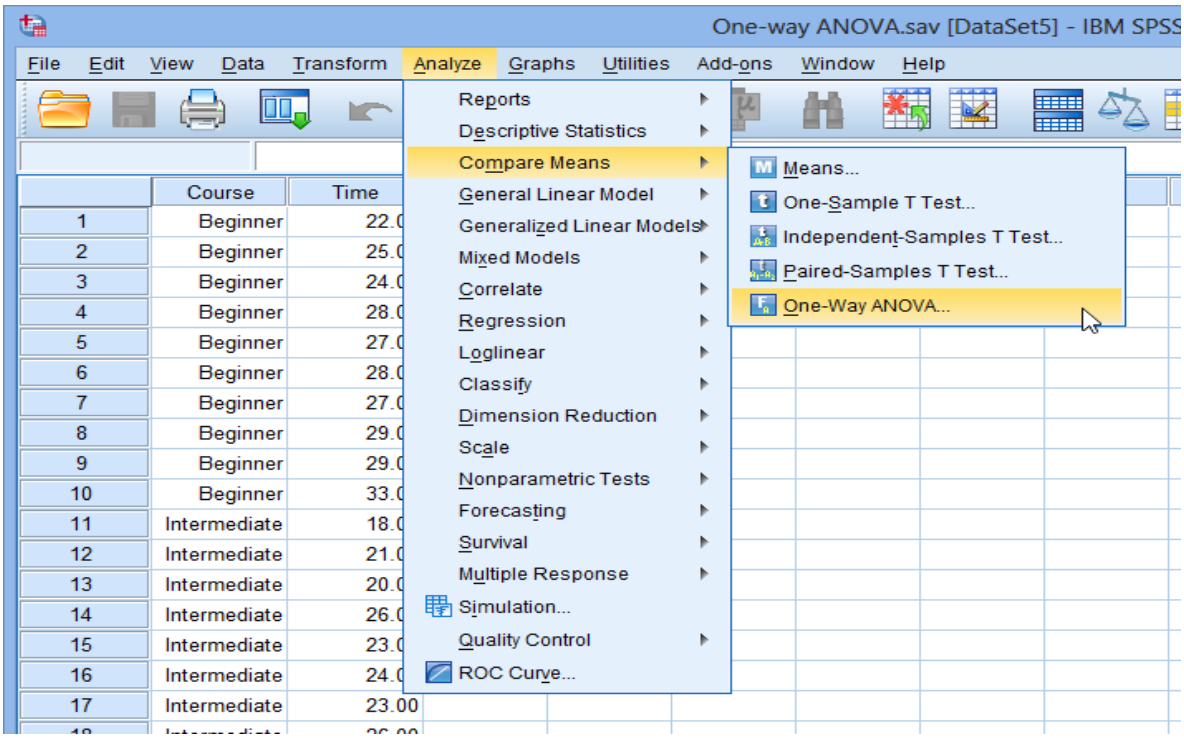
Steps for execution in SPSS

In SPSS Statistics, we separated the groups for analysis by creating a grouping variable called Course (i.e., the independent variable), and assign the values to different courses:

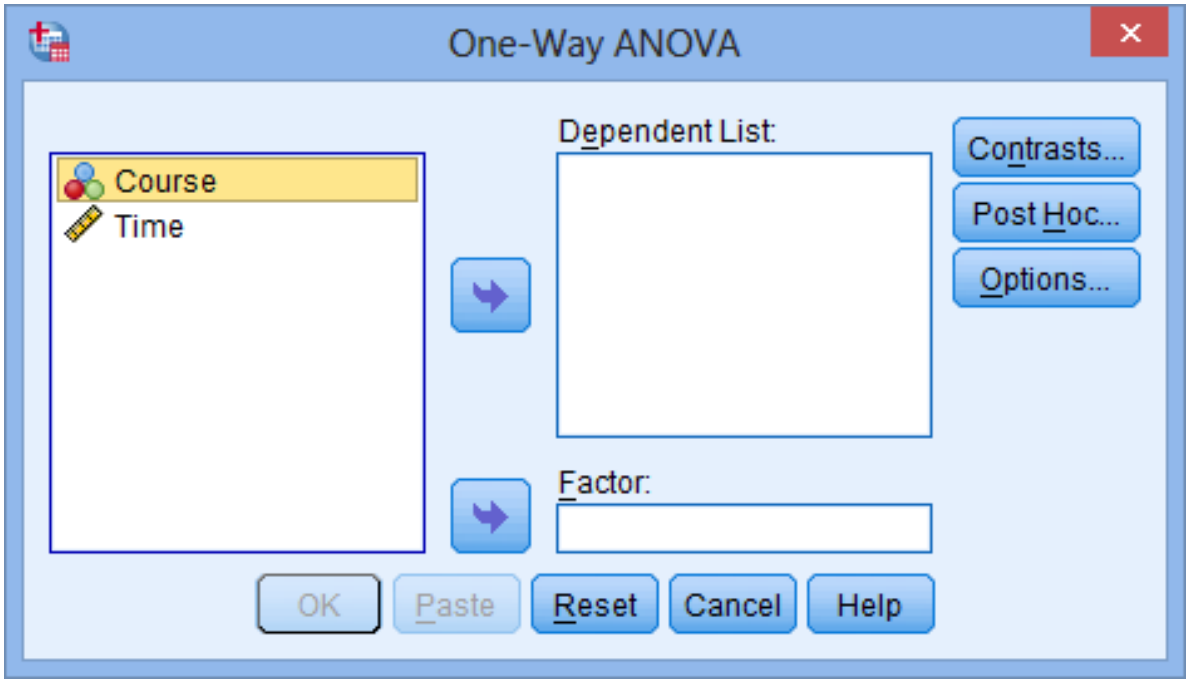
- Beginners course a value of "1"
- The Intermediate course a value of "2"
- The Advanced course a value of "3".

Time to complete the set problem was entered under the variable name Time (i.e., the dependent variable).

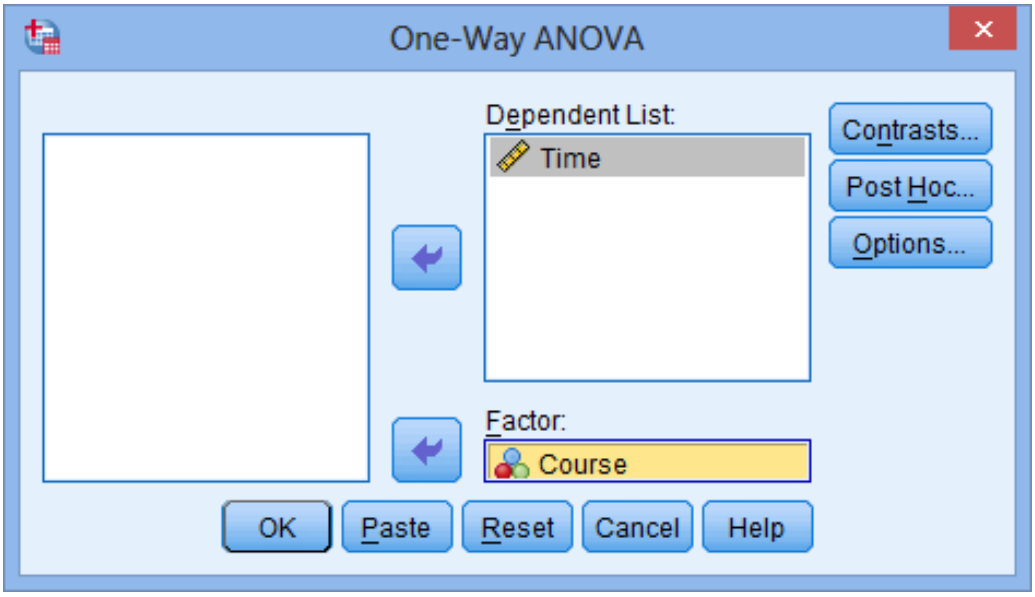
1. Click **Analyze > Compare Means > One-Way ANOVA...** on the top menu, as shown below.



2. You will be presented with the **One-Way ANOVA** dialogue box:

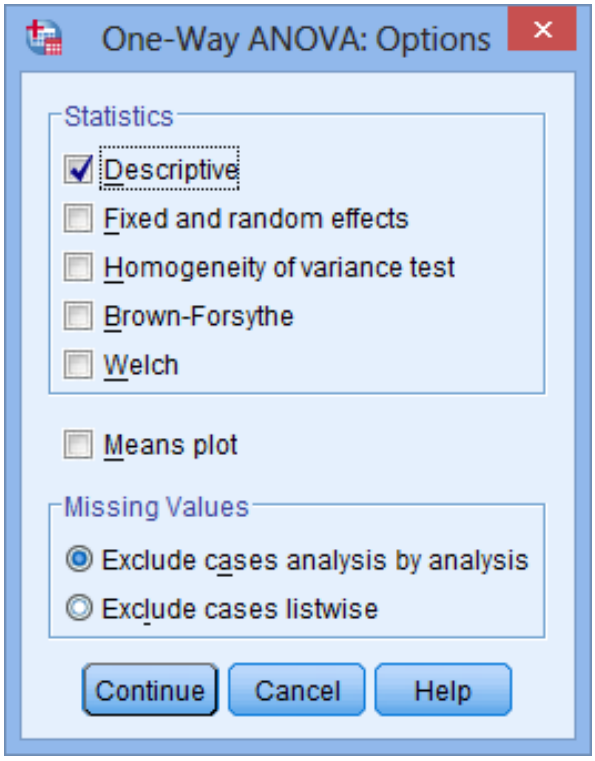


3. Transfer the dependent variable, Time, into **the Dependent List: box** and the independent variable, Course, into the **Factor: box** using the appropriate buttons (or drag-and-drop the variables into the boxes), as shown below:



4. Click on the Continue button.

5. Click on the Options... button > Tick the **Descriptive checkbox** in the –Statistics– area, as shown below:



NOTE: When testing for some of the assumptions of the one-way ANOVA, you will need to tick more of these checkboxes.

6. Click on the Continue button.

7. Click on the OK button.

Output of the One-way ANOVA

Descriptives Table

The descriptives table provides some very useful descriptive statistics, including the mean, standard deviation and 95% confidence intervals for the dependent variable (Time) for each separate group (Beginners, Intermediate and Advanced), as well as when all groups are combined (Total). These figures are useful when you need to describe your data.

Descriptives

Time

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Beginner	10	27.2000	3.04777	.96379	25.0198	29.3802	22.00	33.00
Intermediate	10	23.6000	3.30656	1.04563	21.2346	25.9654	18.00	29.00
Advanced	10	23.4000	3.23866	1.02415	21.0832	25.7168	18.00	29.00
Total	30	24.7333	3.56161	.65026	23.4034	26.0633	18.00	33.00

ANOVA Table

Key columns are F, df (between, within / error), Sig. (p-value).

- If $p < \alpha$ (commonly .05), you reject the null hypothesis and conclude there is a statistically significant difference in means between at least two groups.
- If $p \geq \alpha$, you fail to reject the null (no evidence of difference in group means)

ANOVA

Time

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	91.467	2	45.733	4.467	.021
Within Groups	276.400	27	10.237		
Total	367.867	29			

- **Significant value:** 0.021
- **F – value:** 4.467
- **Degree of Freedom:**
    - **Between the groups:** 2
    - **Within groups:** 27

We can see that **the significance value is 0.021** (i.e., $p = .021$), which is **below 0.05**. and, therefore, **there is a statistically significant difference in the mean length of time to complete the spreadsheet problem between the different courses.**

Conclusion & Interpretation

We Reject the Null Hypothesis and conclude that the type of course affects the average time taken to solve the spreadsheet problem. This means that at least one course leads to faster or slower problem-solving compared to the others.

- One-Way ANOVA is a powerful tool when comparing group means.
- In this case, it showed that training type had a real impact on spreadsheet performance.
- Practical takeaway: Choosing the right training program improves employee productivity.